VLAN Configuration and Trunking

Cisco Packet Tracer Network Project Documentation

1. Project Overview

This project demonstrates how to create and configure two VLANs—IT and HR—across two Cisco switches. Access ports connect end devices to the appropriate VLAN, while an inter-switch trunk link carries traffic for both VLANs. Devices in the same VLAN can communicate across the switches, while IT and HR remain separate broadcast domains.

2. Objectives

• Create VLAN 10 (IT) and VLAN 20 (HR) on both switches.

• Assign end-device access ports to the correct VLAN.

• Configure GigabitEthernet0/1 between S1 and S2 as a trunk.

• Save switch configurations and verify VLAN and trunk status.

• Test connectivity between same-VLAN PCs across the switches.

3. Network Devices and IP Addressing

The topology contains two switches (S1 and S2) and four PCs. Connect S1 GigabitEthernet0/1 to S2 GigabitEthernet0/1. The exact PC access-port numbers were not specified, so connect each PC to an available port within its assigned range.

Switch	PC	IPv4 address	Subnet mask	VLAN
S1	IT-PC1	192.168.10.10	255.255.255.0	10 – IT
S2	IT-PC2	192.168.10.11	255.255.255.0	10 – IT
S1	HR-PC1	192.168.20.10	255.255.255.0	20 – HR
S2	HR-PC2	192.168.20.11	255.255.255.0	20 – HR

4. VLAN and Port Plan

VLAN ID	VLAN name	Network	Access-port range
10	IT	192.168.10.0/24	Fa0/1–Fa0/12
20	HR	192.168.20.0/24	Fa0/13–Fa0/24

5. Switch Configuration

Apply this configuration on both switches. Use hostname S1 on the first switch and hostname S2 on the second.

```
enable
configure terminal
hostname S1
vlan 10
 name IT
vlan 20
 name HR
!
interface range fa0/1-12
 switchport mode access
 switchport access vlan 10
!
```

```
interface range fa0/13-24
 switchport mode access
 switchport access vlan 20
end
copy running-config startup-config
```

On S2, replace the hostname command with hostname S2. The VLAN and access-port commands remain the same.

6. Trunk Configuration

Configure the inter-switch uplink on both switches. The correct command is switchport mode trunk; this also corrects the shortened command in the original S2 notes.

```
enable
configure terminal
interface gigabitEthernet0/1
 switchport mode trunk
end
copy running-config startup-config
```

7. Verification Commands

```
show vlan brief
show interfaces trunk
show running-config
show interfaces gigabitEthernet0/1 switchport
```

Expected results: VLANs 10 and 20 should appear on both switches; access ports should be listed under their assigned VLANs; and GigabitEthernet0/1 should appear as trunking and carry VLANs 10 and 20.

8. Connectivity Testing

From IT-PC1, ping 192.168.10.11 (IT-PC2). From HR-PC1, ping 192.168.20.11 (HR-PC2). Both same-VLAN tests should succeed when cabling and configuration are correct. A ping between an IT PC and an HR PC is expected to fail in this Layer 2 design because no router or Layer 3 switch is configured for inter-VLAN routing.

9. Important Notes

• All PCs use subnet mask 255.255.255.0 (/24). A default gateway is not needed for the same-VLAN tests in this project.

• Connect IT PCs to ports Fa0/1–Fa0/12 and HR PCs to ports Fa0/13–Fa0/24 on each switch.

• Ensure the S1–S2 cable connects GigabitEthernet0/1 on both switches.

• If trunking does not work, check cabling, interface status, and show interfaces trunk output.

• This project does not include DHCP, routing, or inter-VLAN communication.

10. Conclusion

This topology provides hands-on practice with VLAN creation, access-port assignment, trunk configuration, configuration saving, and connectivity verification using Cisco IOS show commands and ping tests.